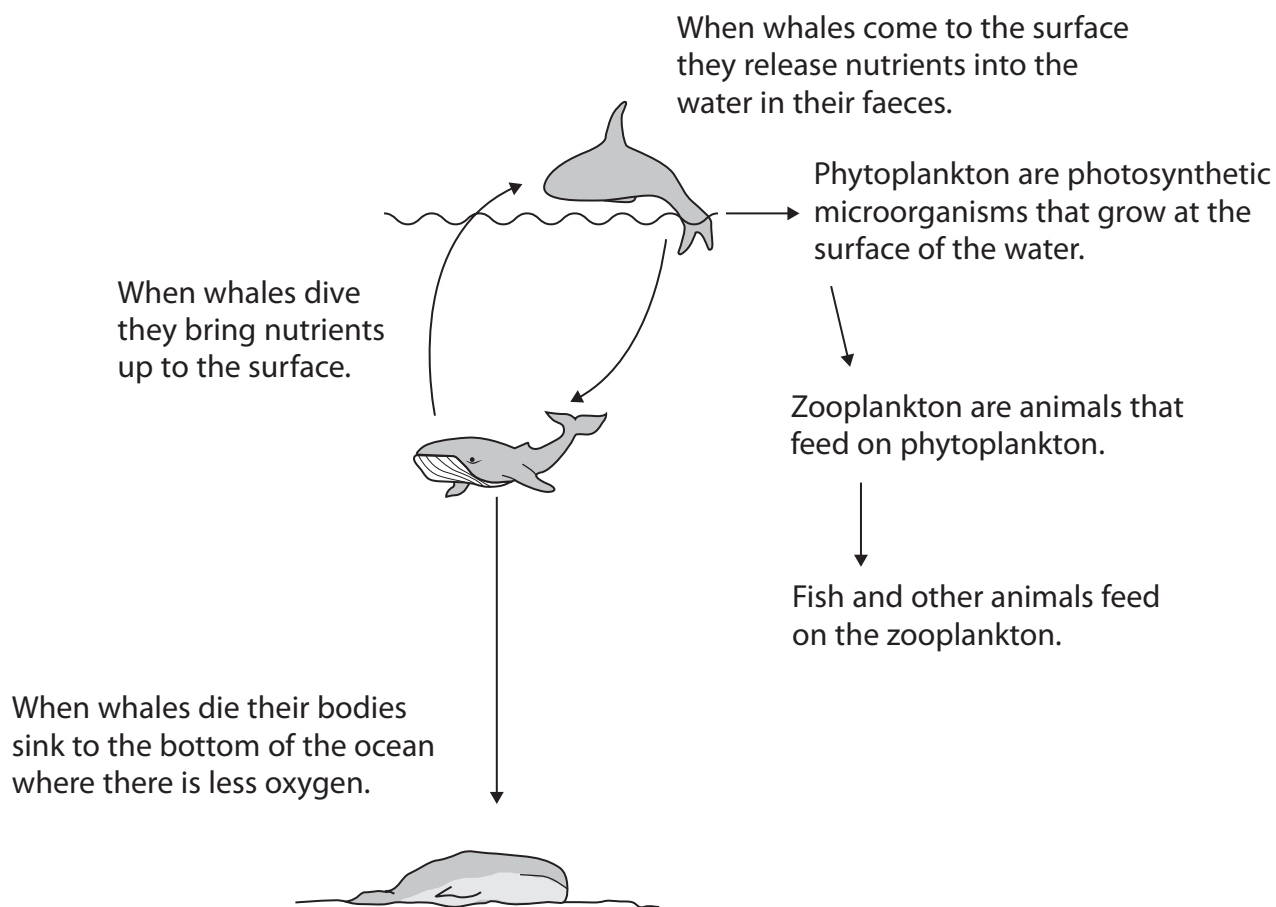


7 Commercial whaling has resulted in a large decrease in the number of whales.

The diagram shows the relationships between whales and some other organisms in the ocean.



(a) Whale faeces add nutrients containing nitrogen to the water.

- (i) There are 345 Right whales in the Gulf of Maine and together they release 15.9 kg of nitrogen per day.

Calculate the mass of nitrogen released by one Right whale in a year.

(1)

Answer kg per year

- (ii) Nitrates contain nitrogen.

Describe the importance of nitrates to plants.

(2)

- (b) There are 1.3 million whales in the oceans.

Scientists believe that if the number of whales could be restored to their original numbers, this would have a significant effect on climate change.

- (i) There were approximately 5 million whales before modern commercial whaling.

Calculate the percentage decrease in the number of whales due to commercial whaling.

(1)

Answer%

- (ii) Calculations have shown that if numbers could be restored, 1.7×10^9 US tons of carbon dioxide could be removed from the atmosphere per year.

A US ton is equivalent to 907 kg.

Calculate the mass of carbon dioxide, in kg, that could be removed from the atmosphere in one year.

Give your answer in standard form.

(1)

Answer kg

- *(iii) Discuss the possible effects of increasing the number of whales on climate change.

Use the information in the diagram and the data given in this question to support your answer.

(6)

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

(Total for Question 7 = 11 marks)